Regression Analysis

Practice problems for the SNU Statistics graduate entrance exam

Part I: 31 original problems, solutions from page 7.

Part II: 22 past exam problems (2020–2026, restated), from page 13.

Notation: unless stated otherwise, $\mathbf{y} = X\beta + \epsilon$ with X of size $n \times (p+1)$ (intercept included) and full column rank, $\epsilon \sim N_n(\mathbf{0}, \sigma^2 I)$, $\hat{\beta} = (X'X)^{-1}X'\mathbf{y}$, $H = X(X'X)^{-1}X' = (h_{ij})$, $\mathbf{e} = \mathbf{y} - X\hat{\beta}$, $SSE = \mathbf{e}'\mathbf{e}$, $MSE = SSE/(n-p-1)$, $(X'X)^{-1} = (c_{ij})$, $J = \mathbf{1}\mathbf{1}'$. In simple regression, $S_{xx} = \sum(x_i - \bar{x})^2$, $S_{xy} = \sum(x_i - \bar{x})(y_i - \bar{y})$. Upper-$\alpha$ points are $t_\alpha(\cdot)$, $F_\alpha(\cdot,\cdot)$, $\chi^2_\alpha(\cdot)$.

Part I. Problems

Problem 1 *Least squares in simple regression* **[10 pts]**

For $y_i = \beta_0 + \beta_1 x_i + \epsilon_i$, $i = 1, \dots, n$:

(a) Derive $\hat{\beta}_1 = S_{xy}/S_{xx}$ and $\hat{\beta}_0 = \bar{y} - \hat{\beta}_1\bar{x}$, and show they minimize $\sum(y_i - \beta_0 - \beta_1 x_i)^2$.

(b) Show $\sum e_i = 0$, $\sum x_i e_i = 0$ and $\sum \hat{y}_i e_i = 0$, and deduce $SST = SSR + SSE$.

(c) Show $R^2 = r_{xy}^2$. If b is the slope of the regression of x on y, show $\hat{\beta}_1 b = r_{xy}^2$.

Problem 2 *Inference in simple regression* **[10 pts]**

Data: $x = (1, 2, 3, 4, 5)$, $y = (2, 4, 5, 4, 6)$.

(a) Fit the line and compute SSE, MSE and R^2.

(b) Give a 95% confidence interval for β_1 ($t_{0.025}(3) = 3.182$) and test $H_0 : \beta_1 = 0$.

(c) Give a 95% prediction interval for a new observation at $x = 6$.

(d) Show that $\mathrm{Cov}(\hat{\beta}_0, \hat{\beta}_1) = -\bar{x}\sigma^2/S_{xx}$ and explain when the two estimates are uncorrelated.

Problem 3 *Regression through the origin* **[10 pts]**

Consider $y_i = \beta x_i + \epsilon_i$, $\epsilon_i \overset{\text{iid}}{\sim} N(0, \sigma^2)$.

(a) Find $\hat{\beta}$, its distribution, and an unbiased estimator of σ^2 (state its degrees of freedom).

(b) Show that $\sum x_i e_i = 0$ but $\sum e_i$ need not be 0.

(c) Explain why $1 - SSE/\sum(y_i - \bar{y})^2$ may be negative here, and give the appropriate decomposition of $\sum y_i^2$.

Problem 4 *Gauss–Markov for the slope* **[10 pts]**

In simple regression with uncorrelated errors of common variance σ^2, let $\tilde{\beta}_1 = \sum c_i y_i$ be linear and unbiased for β_1.

(a) Show that $\sum c_i = 0$ and $\sum c_i x_i = 1$.

(b) Minimize $\mathrm{Var}\,\tilde{\beta}_1 = \sigma^2 \sum c_i^2$ under these constraints and show the minimizer is $c_i = (x_i - \bar{x})/S_{xx}$.

(c) For a design restricted to $x_i \in [-1, 1]$ with n even, which design minimizes $\mathrm{Var}\,\hat{\beta}_1$?

Problem 5 *Expected mean squares* **[10 pts]**

In simple regression:

(a) Show $E(SSR) = \sigma^2 + \beta_1^2 S_{xx}$ and $E(SSE) = (n-2)\sigma^2$.

(b) Show that $F_0 = MSR/MSE$ equals t_0^2, where $t_0 = \hat\beta_1 / \sqrt{MSE/S_{xx}}$.

(c) Identify the noncentrality parameter of the distribution of SSR/σ^2.

Problem 6 *Inference on correlation* **[10 pts]**

(X, Y) is bivariate normal and r is the sample correlation of n pairs.

(a) Show that the t statistic for $\beta_1 = 0$ in the regression of Y on X equals $r\sqrt{n-2}/\sqrt{1-r^2}$.

(b) With $n = 28$ and $r = 0.6$, test $H_0 : \rho = 0$ at 5% ($t_{0.025}(26) = 2.056$).

(c) Using Fisher's z, give an approximate 95% confidence interval for ρ.

Problem 7 *Transformations* **[10 pts]**

(a) Show that $(y^\lambda - 1)/\lambda \to \log y$ as $\lambda \to 0$.

(b) Suppose $\operatorname{Var}(y) = g(\mu)$ with $\mu = Ey$. Using a first-order Taylor expansion, find h such that $\operatorname{Var} h(y)$ is approximately constant when $g(\mu) = \mu$ and when $g(\mu) = \mu^2$.

(c) Explain why, in the Box–Cox procedure, the residual sums of squares are computed for $z^{(\lambda)} = (y^\lambda - 1)/(\lambda \dot y^{\lambda-1})$ ($\dot y$ the geometric mean) rather than for $y^{(\lambda)}$.

Problem 8 *Errors in variables* **[10 pts]**

The true model is $y = \beta_0 + \beta_1 x^* + \epsilon$, but $x = x^* + \delta$ is observed, where x^*, δ, ϵ are independent with $\operatorname{Var} x^* = \sigma_*^2$, $\operatorname{Var}\delta = \sigma_\delta^2$.

(a) Find the probability limit of the least squares slope of y on x.

(b) If σ_δ^2 is known (from replicate measurements), propose a consistent estimator of β_1.

(c) In Berkson's model, $x^* = x + \delta$ with x set by the experimenter and δ independent of x. Is least squares unbiased?

Problem 9 *The hat matrix* **[10 pts]**

(a) Show that H and $I - H$ are symmetric idempotent with traces $p+1$ and $n-p-1$.

(b) Show $\mathbf{e} = (I-H)\epsilon$, $\operatorname{Var}\mathbf{e} = \sigma^2(I-H)$ and $\operatorname{Cov}(\hat\beta, \mathbf{e}) = 0$.

(c) Show $\frac{1}{n} \le h_{ii} \le 1$ (with an intercept) and that $h_{ii} = 1$ implies $e_i = 0$.

(d) In simple regression show $h_{ii} = \frac{1}{n} + \frac{(x_i - \bar x)^2}{S_{xx}}$.

Problem 10 *Quadratic forms* **[10 pts]**

Let $\mathbf{y} \sim N(\mu, \Sigma)$ and A symmetric.

(a) Show $E(\mathbf{y}'A\mathbf{y}) = \operatorname{tr}(A\Sigma) + \mu' A \mu$.

(b) Use (a) to compute $E(SST)$, $E(SSR)$ and $E(SSE)$ in the multiple regression model.

(c) Show that $\mathbf{y}'(H - J/n)\mathbf{y}$ and $\mathbf{y}'(I - H)\mathbf{y}$ are independent and give their distributions.

Problem 11 *Overall F test* **[10 pts]**

(a) Show that $F_0 = MSR/MSE$ has the noncentral distribution $F(p, n-p-1, \lambda)$ with $\lambda = \beta' X'(I - J/n)X\beta/(2\sigma^2)$.

(b) Show that $\lambda = 0$ iff $\beta_1 = \cdots = \beta_p = 0$.

(c) Express F_0 in terms of R^2.

Problem 12 *Partial regression* **[10 pts]**

Let $X = (X_1, X_2)$ and $H_1 = X_1(X_1'X_1)^{-1}X_1'$.

(a) Show $\hat{\beta}_2 = \{X_2'(I - H_1)X_2\}^{-1}X_2'(I - H_1)\mathbf{y}$.

(b) Show that the residuals from regressing $(I - H_1)\mathbf{y}$ on $(I - H_1)X_2$ equal the full-model residuals.

(c) If $X_1'X_2 = 0$, show that $\hat{\beta}_1$ is the same in the full model and in the model with X_1 alone, and that SSR splits into two parts.

Problem 13 *Restricted least squares* **[10 pts]**

Data with X having rows $(1, x_{i1}, x_{i2})$:

x_1	1	2	3	4	5	6
x_2	0	1	1	3	2	4
y	3	6	6	10	9	13

Here $X'X = \begin{pmatrix} 6 & 21 & 11 \\ 21 & 91 & 51 \\ 11 & 51 & 31 \end{pmatrix}$, $(X'X)^{-1} = \frac{1}{200}\begin{pmatrix} 220 & -90 & 70 \\ -90 & 65 & -75 \\ 70 & -75 & 105 \end{pmatrix}$, $X'\mathbf{y} = (47, 196, 112)'$.

(a) Compute $\hat{\beta}$ and SSE.

(b) Test $H_0 : \beta_1 = \beta_2$ at 5% by the formula $Q = (C\hat{\beta})'\{C(X'X)^{-1}C'\}^{-1}C\hat{\beta}$ ($F_{0.05}(1,3) = 10.13$).

(c) Verify the result by fitting the reduced model.

Problem 14 *Partial and sequential F* **[10 pts]**

(a) Show $SS(\hat{\beta}_j \mid \text{all others}) = \hat{\beta}_j^2/c_{jj}$, so the partial F test equals the squared t test.

(b) Show that the sequential sums of squares $SS(\hat{\beta}_1 \mid \hat{\beta}_0)$, $SS(\hat{\beta}_2 \mid \hat{\beta}_0, \hat{\beta}_1)$, … add up to SSR.

(c) Show that, if the centred regressors are mutually orthogonal, the sequential sums of squares do not depend on the order.

Problem 15 *Simultaneous inference* **[10 pts]**

(a) Show that $(\hat{\beta} - \beta)'X'X(\hat{\beta} - \beta)/\{(p+1)MSE\} \sim F(p+1, n-p-1)$.

(b) Derive Scheffé's simultaneous intervals for all $\mathbf{a}'\beta$ from (a).

(c) For $p+1=3$ and $n-p-1=20$, compare the multipliers of single, Bonferroni and Scheffé intervals for the three coefficients ($t_{0.025}(20)=2.086$, $t_{0.05/6}(20)=2.613$, $F_{0.05}(3,20)=3.098$).

Problem 16 *Comparing two regressions* **[10 pts]**

Two groups follow $y_{ij}=\beta_{0i}+\beta_{1i}x_{ij}+\epsilon_{ij}$, $i=1,2$, $j=1,\dots,n_i$, with a common σ^2.

(a) Write a single regression with an indicator d (for group 2) and its interaction with x that reproduces the two separate lines.

(b) Show that its SSE equals SSE_1+SSE_2 of the separate fits.

(c) Give the F tests for identical lines and for parallel lines, with degrees of freedom.

Problem 17 *Deletion formulas* **[10 pts]**

Let $\hat\beta_{(i)}$ and $s^2_{(i)}$ be computed without observation i.

(a) Using $(A-\mathbf{x}\mathbf{x}')^{-1}=A^{-1}+A^{-1}\mathbf{x}\mathbf{x}'A^{-1}/(1-\mathbf{x}'A^{-1}\mathbf{x})$, prove $\hat\beta_{(i)}=\hat\beta-(X'X)^{-1}\mathbf{x}_i e_i/(1-h_{ii})$.

(b) Prove $(n-p-2)s^2_{(i)}=(n-p-1)s^2-e_i^2/(1-h_{ii})$.

(c) Show Cook's distance satisfies $D_i=\frac{r_i^2}{p+1}\cdot\frac{h_{ii}}{1-h_{ii}}$, with $r_i=e_i/(s\sqrt{1-h_{ii}})$.

Problem 18 *Studentized residuals and PRESS* **[10 pts]**

(a) Show that $r_i^*=e_i/(s_{(i)}\sqrt{1-h_{ii}})\sim t(n-p-2)$.

(b) Show $r_i^*=r_i\sqrt{(n-p-2)/(n-p-1-r_i^2)}$, and deduce that $r_i^2\le n-p-1$.

(c) Show that the leave-one-out cross-validation error is PRESS $=\sum_i\{e_i/(1-h_{ii})\}^2$.

Problem 19 *Omitted variables* **[10 pts]**

The true model is $y=\beta_0+\beta_1x_1+\beta_2x_2+\epsilon$, but y is regressed on x_1 alone.

(a) Show $E\tilde\beta_1=\beta_1+\beta_2 S_{12}/S_{11}$, with $S_{12}=\sum(x_{i1}-\bar x_1)(x_{i2}-\bar x_2)$.

(b) Compare $\operatorname{Var}\tilde\beta_1$ with $\operatorname{Var}\hat\beta_1$ in the full model, in terms of the sample correlation r_{12}.

(c) Show that $\operatorname{MSE}(\tilde\beta_1)\le\operatorname{Var}(\hat\beta_1)$ iff $\beta_2^2\le\operatorname{Var}(\hat\beta_2)$.

Problem 20 *Selection criteria* **[10 pts]**

(a) Show that R^2 cannot decrease when a regressor is added.

(b) Show that R^2_a increases when a regressor is added iff its partial F statistic exceeds 1.

(c) Show that $E(C_k)\approx k+1$ when the k-variable model is correct, and that $C_p=p+1$ for the full model.

Problem 21 *Weighted least squares* **[10 pts]**

Consider $y_i = \beta x_i + \epsilon_i$ with $x_i > 0$ and independent $\epsilon_i \sim N(0, \sigma^2 x_i)$.

(a) Find the GLS estimator of β and its variance.

(b) Find the variance of the OLS estimator $\sum x_i y_i / \sum x_i^2$ and verify that it is not smaller.

(c) Give an unbiased estimator of σ^2.

Problem 22 *When OLS is GLS* **[10 pts]**

Let $\operatorname{Var}\epsilon = \sigma^2 V$ with $V = (1-\rho)I + \rho J$, $-\frac{1}{n-1} < \rho < 1$, and let X contain the intercept column.

(a) Show that $V^{-1} = \frac{1}{1-\rho}\big(I - \frac{\rho}{1-\rho+n\rho}J\big)$.

(b) Show that the GLS and OLS estimators of β coincide.

(c) Is the usual MSE unbiased for σ^2 in this model?

Problem 23 *Indicator variables* **[10 pts]**

(a) A factor has three levels. Explain why using three indicators plus an intercept makes $X'X$ singular.

(b) With baseline coding, interpret the coefficients of $E(y) = \beta_0 + \beta_1 x + \beta_2 d_1 + \beta_3 d_2$.

(c) How would you test that the three levels have the same intercept? Give the degrees of freedom.

Problem 24 *Piecewise linear regression* **[10 pts]**

Consider $E(y) = \beta_0 + \beta_1 x + \beta_2 (x - c)_+$ with known c.

(a) Show that the mean function is continuous at c and find the slopes on each side.

(b) Modify the model to allow a jump at c.

(c) How would you test whether the slope changes at c?

Problem 25 *Ridge regression* **[10 pts]**

Let $\hat\beta(k) = (X'X + kI)^{-1}X'\mathbf{y}$, with X standardized and $X'X = P\Lambda P'$.

(a) Show that $\hat\beta(k)$ minimizes $\|\mathbf{y} - X\beta\|^2 + k\|\beta\|^2$ and is the OLS estimator for the augmented data $\binom{\mathbf{y}}{\mathbf{0}}$, $\binom{X}{\sqrt{k}I}$.

(b) Find $E\hat\beta(k)$ and $\operatorname{Var}\hat\beta(k)$.

(c) Show that $E\|\hat\beta(k) - \beta\|^2 < E\|\hat\beta - \beta\|^2$ for some $k > 0$.

(d) If $X'X = I$, find the k minimizing $E\|\hat\beta(k) - \beta\|^2$.

Problem 26 *Variance inflation* **[10 pts]**

(a) Show that $c_{jj} = (1 - R_j^2)^{-1}$ for standardized regressors, where R_j^2 is from regressing x_j on the others.

(b) Show $\sum_j \mathrm{VIF}_j = \sum_j 1/\lambda_j$, with λ_j the eigenvalues of $X'X$.

(c) For two regressors with correlation 0.95, find the VIFs and the factor by which confidence intervals widen.

Problem 27 *Principal component regression* **[10 pts]**

With $X'X = \sum_k l_k \mathbf{e}_k \mathbf{e}_k'$ ($l_1 \ge \cdots \ge l_p$), the PCR estimator with r components is

$$\tilde{\beta}_r = \sum_{k \le r} l_k^{-1} \mathbf{e}_k \mathbf{e}_k' X' \mathbf{y}.$$

(a) Find the bias and covariance of $\tilde{\beta}_r$.

(b) Show $E\|\tilde{\beta}_r - \beta\|^2 < E\|\hat{\beta} - \beta\|^2$ iff $\sum_{k>r} (\mathbf{e}_k'\beta)^2 < \sigma^2 \sum_{k>r} 1/l_k$.

(c) Contrast PCR with ridge in terms of the shrinkage factors applied to the canonical coefficients.

Problem 28 *Logistic regression: estimation* **[10 pts]**

For $P(y_i = 1) = \pi_i = e^{\mathbf{x}_i'\beta}/(1 + e^{\mathbf{x}_i'\beta})$:

(a) Derive the score and Hessian, and show that the log-likelihood is concave.

(b) Show that a Newton–Raphson step is a weighted least squares regression of a working response.

(c) Interpret e^{β_j}.

Problem 29 *Logistic regression:* 2×2 *tables* **[10 pts]**

(a) For a binary x, show that the MLEs of the logistic model give $e^{\hat{\beta}_1} = \frac{n_{11} n_{00}}{n_{10} n_{01}}$.

(b) Show that the odds ratio is the same whether computed from $P(y \mid x)$ or $P(x \mid y)$, and explain why this matters for case–control studies.

(c) With $n_{11} = 20, n_{10} = 80, n_{01} = 10, n_{00} = 90$, compute $\widehat{OR}$ and an approximate 95% interval, using $\mathrm{se}(\log \widehat{OR}) = \sqrt{\sum 1/n_{xy}}$.

Problem 30 *Separation* **[10 pts]**

Data: $x = (1, 2, 3, 4)$, $y = (0, 0, 1, 1)$, with model $\operatorname{logit} P(y = 1) = \beta_0 + \beta_1 x$.

(a) Show that the log-likelihood along $\beta = t(-2.5, 1)$ increases to 0 as $t \to \infty$.

(b) Conclude that the MLE does not exist.

(c) Can this happen in ordinary least squares?

Problem 31 *Lack of fit and replication* **[10 pts]**

At each of m distinct x-values x_j there are n_j replicates y_{jk}, $n = \sum n_j$, and the straight-line model is fitted.

(a) Show $SSE = SS_{PE} + SS_{LOF}$, where $SS_{PE} = \sum_j \sum_k (y_{jk} - \bar{y}_j)^2$ and $SS_{LOF} = \sum_j n_j (\bar{y}_j - \hat{y}_j)^2$.

(b) Show that $F = \frac{SS_{LOF}/(m-2)}{SS_{PE}/(n-m)} \sim F(m-2, n-m)$ if the line is correct.

(c) Why can lack of fit not be tested this way without replication?

Part I. Solutions

Solution 1

(a) The normal equations $\sum(y_i - \beta_0 - \beta_1 x_i) = 0$ and $\sum x_i(y_i - \beta_0 - \beta_1 x_i) = 0$ give $\beta_0 = \bar{y} - \beta_1 \bar{x}$ and then $\beta_1 S_{xx} = S_{xy}$. The objective is a convex quadratic with Hessian $2\begin{pmatrix} n & \sum x_i \\ \sum x_i & \sum x_i^2 \end{pmatrix}$, which has determinant $4nS_{xx} > 0$, so the stationary point is the global minimum.

(b) The first two identities are the normal equations, and $\sum \hat{y}_i e_i = \hat{\beta}_0 \sum e_i + \hat{\beta}_1 \sum x_i e_i = 0$. In $\sum(y_i - \bar{y})^2 = \sum\{e_i + (\hat{y}_i - \bar{y})\}^2$ the cross term is $2\sum e_i \hat{y}_i - 2\bar{y}\sum e_i = 0$.

(c) $SSR = \sum \hat{\beta}_1^2 (x_i - \bar{x})^2 = S_{xy}^2 / S_{xx}$, so $R^2 = S_{xy}^2/(S_{xx}S_{yy}) = r_{xy}^2$. Also $b = S_{xy}/S_{yy}$, so $\hat{\beta}_1 b = S_{xy}^2/(S_{xx}S_{yy}) = r_{xy}^2$.

Solution 2

(a) $\bar{x} = 3$, $\bar{y} = 4.2$, $S_{xx} = 10$, $S_{xy} = 8$, so $\hat{\beta}_1 = 0.8$ and $\hat{\beta}_0 = 1.8$. $SST = 8.8$, $SSR = 0.8^2 \cdot 10 = 6.4$, $SSE = 2.4$, $MSE = 0.8$, $R^2 = 0.727$.

(b) $\mathrm{se}(\hat{\beta}_1) = \sqrt{0.8/10} = 0.283$; the interval is $0.8 \pm 3.182 \cdot 0.283 = (-0.10, 1.70)$. $t_0 = 2.83 < 3.182$: do not reject $\beta_1 = 0$ (consistent with the interval containing 0).

(c) $\hat{y} = 6.6$, and the prediction interval is $6.6 \pm 3.182\sqrt{0.8(1 + \frac{1}{5} + \frac{9}{10})} = 6.6 \pm 4.12$.

(d) $\mathrm{Cov}(\hat{\beta}_0, \hat{\beta}_1) = \mathrm{Cov}(\bar{y} - \hat{\beta}_1\bar{x}, \hat{\beta}_1) = \mathrm{Cov}(\bar{y}, \hat{\beta}_1) - \bar{x}\,\mathrm{Var}\,\hat{\beta}_1 = 0 - \bar{x}\sigma^2/S_{xx}$, using $\mathrm{Cov}(\bar{y}, \sum k_i y_i) = \frac{\sigma^2}{n}\sum k_i = 0$. They are uncorrelated iff $\bar{x} = 0$, e.g. after centring x.

Solution 3

(a) Minimizing $\sum(y_i - \beta x_i)^2$ gives $\hat{\beta} = \sum x_i y_i / \sum x_i^2 \sim N(\beta, \sigma^2/\sum x_i^2)$. $X = \mathbf{x}$ has rank 1, so $SSE/\sigma^2 \sim \chi^2(n-1)$ and $MSE = SSE/(n-1)$ is unbiased.

(b) The only normal equation is $\sum x_i(y_i - \hat{\beta}x_i) = 0$. Nothing forces $\sum e_i = 0$, since $\mathbf{1}$ is not in the column space; e.g. $x = (1, 2)$, $y = (1, 1)$ gives $\hat{\beta} = 0.6$ and $\mathbf{e} = (0.4, -0.2)$.

(c) Without an intercept, $\sum(y_i - \bar{y})^2 = SSE + \sum(\hat{y}_i - \bar{y})^2$ fails, and SSE can exceed $\sum(y_i - \bar{y})^2$ (a line through the origin can fit worse than the horizontal line $\bar{y}$). The valid decomposition is the uncorrected one, $\sum y_i^2 = \sum \hat{y}_i^2 + \sum e_i^2$ (since $\hat{\mathbf{y}} \perp \mathbf{e}$), with the uncentred $R^2 = \sum \hat{y}_i^2 / \sum y_i^2$.

Solution 4

(a) $E\sum c_i y_i = \beta_0 \sum c_i + \beta_1 \sum c_i x_i = \beta_1$ for all β_0, β_1 forces $\sum c_i = 0$ and $\sum c_i x_i = 1$.

(b) Let $k_i = (x_i - \bar{x})/S_{xx}$ and $d_i = c_i - k_i$. Then $\sum d_i = 0$ and $\sum d_i x_i = 0$, so $\sum k_i d_i = \frac{1}{S_{xx}}\sum(x_i - \bar{x})d_i = 0$ and $\sum c_i^2 = \sum k_i^2 + \sum d_i^2 \geq \sum k_i^2$, with equality iff $d = 0$.

(c) $\mathrm{Var}\,\hat{\beta}_1 = \sigma^2/S_{xx}$, and $S_{xx} \leq n$ on $[-1, 1]$ with equality iff half the points are at -1 and half at $+1$. (This design cannot detect curvature.)

Solution 5

(a) $E(SSR) = S_{xx}E\hat{\beta}_1^2 = S_{xx}(\sigma^2/S_{xx} + \beta_1^2)$. $E(SST) = E\sum\{\beta_1(x_i - \bar{x}) + \epsilon_i - \bar{\epsilon}\}^2 = \beta_1^2 S_{xx} + (n-1)\sigma^2$; subtract.

(b) $F_0 = \hat{\beta}_1^2 S_{xx}/MSE = t_0^2$.

(c) $SSR/\sigma^2 = (\hat{\beta}_1\sqrt{S_{xx}}/\sigma)^2$ with $\hat{\beta}_1\sqrt{S_{xx}}/\sigma \sim N(\beta_1\sqrt{S_{xx}}/\sigma, 1)$, so it is $\chi^2(1, \lambda)$ with $\lambda = \beta_1^2 S_{xx}/(2\sigma^2)$ (half-convention).

Solution 6

(a) $SSR = r^2 SST$ and $SSE = (1 - r^2)SST$, so $t_0^2 = SSR/(SSE/(n-2)) = (n-2)r^2/(1-r^2)$.

(b) $t_0 = 0.6\sqrt{26}/0.8 = 3.82 > 2.056$: reject $\rho = 0$.

(c) $z = \tanh^{-1} 0.6 = 0.693$, half-width $1.96/\sqrt{25} = 0.392$; back-transforming $(0.301, 1.085)$ gives $(0.29, 0.80)$.

Solution 7

(a) $y^\lambda = e^{\lambda \log y} = 1 + \lambda \log y + O(\lambda^2)$.

(b) $\operatorname{Var} h(y) \approx h'(\mu)^2 g(\mu)$, so take $h'(\mu) \propto g(\mu)^{-1/2}$: $g = \mu$ gives $h = \sqrt{y}$, and $g = \mu^2$ gives $h = \log y$.

(c) $y^{(\lambda)}$ has different units for different λ, so its SSE values are not comparable. The density of y includes the Jacobian $\prod y_i^{\lambda-1} = \dot{y}^{n(\lambda-1)}$; dividing by $\dot{y}^{\lambda-1}$ absorbs it, and then the profile log-likelihood is $-\frac{n}{2} \log SSE(\lambda)$+const. So minimizing SSE of $z^{(\lambda)}$ is maximum likelihood.

Solution 8

(a) $\hat{\beta}_1 \xrightarrow{p} \operatorname{Cov}(x, y)/\operatorname{Var} x = \beta_1 \sigma_*^2/(\sigma_*^2 + \sigma_\delta^2)$: attenuation towards 0.

(b) $\hat{\beta}_1/\hat{\kappa}$ with the reliability $\hat{\kappa} = (s_x^2 - \sigma_\delta^2)/s_x^2$, which converges to $\sigma_*^2/(\sigma_*^2 + \sigma_\delta^2)$; equivalently $S_{xy}/(S_{xx} - (n-1)\sigma_\delta^2)$.

(c) Yes: $y = \beta_0 + \beta_1 x + (\epsilon + \beta_1 \delta)$, and the new error has mean 0 and is independent of the fixed x; only the error variance increases.

Solution 9

(a) $H' = H$, $H^2 = H$, $\operatorname{tr} H = \operatorname{tr}\{(X'X)^{-1}X'X\} = p + 1$; then $(I - H)^2 = I - H$ and $\operatorname{tr}(I - H) = n - p - 1$.

(b) $\mathbf{e} = (I - H)(X\beta + \epsilon) = (I - H)\epsilon$, so $\operatorname{Var} \mathbf{e} = \sigma^2(I - H)$, and $\operatorname{Cov}(\hat{\beta}, \mathbf{e}) = (X'X)^{-1}X'\sigma^2(I - H) = 0$.

(c) $h_{ii} = \sum_j h_{ij}^2 \geq h_{ii}^2$ gives $h_{ii} \leq 1$. With an intercept, $H = J/n + \tilde{H}$ with $\tilde{H}$ the projection onto the centred regressors, so $h_{ii} \geq 1/n$. If $h_{ii} = 1$, then $\sum_{j \neq i} h_{ij}^2 = 0$, so $\hat{y}_i = y_i$ and $e_i = 0$ (also $\operatorname{Var} e_i = 0$).

(d) $\tilde{H} = \tilde{\mathbf{x}}\tilde{\mathbf{x}}'/S_{xx}$ with $\tilde{\mathbf{x}} = (x_i - \bar{x})_i$, so $h_{ii} = \frac{1}{n} + (x_i - \bar{x})^2/S_{xx}$.

Solution 10

(a) $E\mathbf{y}'A\mathbf{y} = E \operatorname{tr}(A\mathbf{y}\mathbf{y}') = \operatorname{tr}\{A(\Sigma + \mu\mu')\}$.

(b) With $\mu = X\beta$, $\Sigma = \sigma^2 I$: $SSE = \mathbf{y}'(I - H)\mathbf{y}$ gives $(n - p - 1)\sigma^2$, since $(I - H)X = 0$. $SSR = \mathbf{y}'(H - J/n)\mathbf{y}$ gives $p\sigma^2 + \beta' X'(I - J/n)X\beta$, and $SST = \mathbf{y}'(I - J/n)\mathbf{y}$ gives $(n-1)\sigma^2 + \beta' X'(I - J/n)X\beta$.

(c) Both matrices are symmetric idempotent (using $HJ = J$) with $(H - J/n)(I - H) = 0$, so the forms are independent, $SSR/\sigma^2 \sim \chi^2(p, \lambda)$ and $SSE/\sigma^2 \sim \chi^2(n - p - 1)$.

Solution 11

(a) Follows from Solution 10(c) and the definition of noncentral F.

(b) $2\sigma^2\lambda = \|(I - J/n)X\beta\|^2$, the squared length of the centred mean vector $\sum_{j \geq 1} \beta_j(\mathbf{x}_j - \bar{x}_j \mathbf{1})$. The centred columns are linearly independent (full rank), so this is 0 iff $\beta_1 = \cdots = \beta_p = 0$.

(c) $F_0 = \dfrac{R^2/p}{(1 - R^2)/(n - p - 1)}$, dividing SSR and SSE by SST.

Solution 12

(a) Multiply the normal equations $\begin{pmatrix} X_2'X_2 & X_2'X_1 \\ X_1'X_2 & X_1'X_1 \end{pmatrix} \binom{\hat{\beta}_2}{\hat{\beta}_1} = \binom{X_2'\mathbf{y}}{X_1'\mathbf{y}}$ by $\begin{pmatrix} I & -X_2'X_1(X_1'X_1)^{-1} \\ 0 & I \end{pmatrix}$. The first row becomes $X_2'(I - H_1)X_2\hat{\beta}_2 = X_2'(I - H_1)\mathbf{y}$.

(b) The full residual $\mathbf{e}$ is orthogonal to X_1, so $\mathbf{e} = (I - H_1)\mathbf{e} = (I - H_1)\mathbf{y} - (I - H_1)X_2\hat{\beta}_2$, which is the residual of the second regression by (a).

(c) Then $X'X$ is block diagonal, so $\hat\beta_1 = (X_1'X_1)^{-1}X_1'\mathbf{y}$ in both models, $H = H_1 + H_2$, and SS due to the regressors splits as $\mathbf{y}'H_1\mathbf{y} + \mathbf{y}'H_2\mathbf{y}$ (centring both blocks when there is an intercept).

Solution 13

(a) $\hat\beta = \frac{1}{200}(220\cdot 47 - 90\cdot 196 + 70\cdot 112,\ -90\cdot 47 + 65\cdot 196 - 75\cdot 112,\ 70\cdot 47 - 75\cdot 196 + 105\cdot 112)' = (2.7, 0.55, 1.75)'$. $\mathbf{y}'\mathbf{y} = 431$ and $\hat\beta' X'\mathbf{y} = 2.7\cdot 47 + 0.55\cdot 196 + 1.75\cdot 112 = 430.7$, so $SSE = 0.3$ and $MSE = 0.1$.

(b) $C = (0, 1, -1)$: $C\hat\beta = -1.2$ and $C(X'X)^{-1}C' = (65 + 105 + 150)/200 = 1.6$, so $Q = 1.44/1.6 = 0.9$ and $F_0 = 0.9/0.1 = 9.0 < 10.13$. Do not reject.

(c) Under H_0, $y = \beta_0 + \beta_1(x_1 + x_2) + \epsilon$. With $z = x_1 + x_2 = (1, 3, 4, 7, 7, 10)$ the fit is $\hat y = 2.1 + 1.075z$ with $SSE(R) = 1.2$. So $SSE(R) - SSE = 0.9 = Q$, as it must be.

Solution 14

(a) With $C = \mathbf{e}_j'$, Theorem (restricted LS) gives $SSE(R) - SSE = \hat\beta_j^2/c_{jj}$, and the partial F is $\hat\beta_j^2/(c_{jj}MSE) = t_0^2$.

(b) The sum telescopes: $\sum_{j=1}^{p}\{SSR(x_1,\dots,x_j) - SSR(x_1,\dots,x_{j-1})\} = SSR(x_1,\dots,x_p) - 0$.

(c) With orthogonal centred regressors, $H - J/n = \sum_j \tilde{\mathbf{x}}_j\tilde{\mathbf{x}}_j'/\|\tilde{\mathbf{x}}_j\|^2$, so adding x_j always contributes $(\tilde{\mathbf{x}}_j'\mathbf{y})^2/\|\tilde{\mathbf{x}}_j\|^2$, regardless of what is already in the model.

Solution 15

(a) $\hat\beta - \beta = (X'X)^{-1}X'\epsilon$, so $(\hat\beta - \beta)'X'X(\hat\beta - \beta) = \epsilon' H\epsilon \sim \sigma^2\chi^2(p+1)$, independent of $SSE = \epsilon'(I - H)\epsilon$.

(b) By Cauchy–Schwarz, $\max_{\mathbf{a}} \frac{\{\mathbf{a}'(\hat\beta - \beta)\}^2}{\mathbf{a}'(X'X)^{-1}\mathbf{a}} = (\hat\beta - \beta)'X'X(\hat\beta - \beta)$. So, with probability $1 - \alpha$, for all $\mathbf{a}$, $|\mathbf{a}'(\hat\beta - \beta)| \le \sqrt{(p+1)F_\alpha(p+1, n-p-1)}\sqrt{\mathbf{a}'(X'X)^{-1}\mathbf{a}\, MSE}$.

(c) The multipliers are 2.086 (individual), 2.613 (Bonferroni) and $\sqrt{3\cdot 3.098} = 3.049$ (Scheffé). Bonferroni is better for three prespecified coefficients; Scheffé covers all combinations.

Solution 16

(a) $E(y) = \beta_{01} + \beta_{11}x + \gamma_0 d + \gamma_1 xd$ with $\gamma_0 = \beta_{02} - \beta_{01}$ and $\gamma_1 = \beta_{12} - \beta_{11}$.

(b) The design is a reparametrization of the block-diagonal design $\mathrm{diag}\{(\mathbf{1}, \mathbf{x}_1), (\mathbf{1}, \mathbf{x}_2)\}$: both span the same column space. The fitted values, and hence SSE, are those of the separate fits.

(c) Identical lines: $H_0 : \gamma_0 = \gamma_1 = 0$, with $F = \frac{\{SSE(R) - SSE(F)\}/2}{SSE(F)/(n_1+n_2-4)} \sim F(2, n_1 + n_2 - 4)$. Parallel lines: $H_0 : \gamma_1 = 0$, with a t test on $n_1 + n_2 - 4$ degrees of freedom (equivalently $F(1, n_1 + n_2 - 4)$).

Solution 17

(a) $X_{(i)}'X_{(i)} = A - \mathbf{x}_i\mathbf{x}_i'$ and $X_{(i)}'\mathbf{y}_{(i)} = X'\mathbf{y} - \mathbf{x}_i y_i$, with $A = X'X$. Then

$$\hat\beta_{(i)} = \Big(A^{-1} + \frac{A^{-1}\mathbf{x}_i\mathbf{x}_i'A^{-1}}{1 - h_{ii}}\Big)(X'\mathbf{y} - \mathbf{x}_i y_i) = \hat\beta - A^{-1}\mathbf{x}_i y_i + \frac{A^{-1}\mathbf{x}_i(\hat y_i - h_{ii}y_i)}{1 - h_{ii}} = \hat\beta - \frac{A^{-1}\mathbf{x}_i e_i}{1 - h_{ii}}.$$

(b) For $j \ne i$, the new residual is $e_j + h_{ij}e_i/(1 - h_{ii})$. Use $\sum_{j\ne i} h_{ij}e_j = -h_{ii}e_i$ (because $H\mathbf{e} = \mathbf{0}$) and $\sum_{j\ne i} h_{ij}^2 = h_{ii} - h_{ii}^2$:

$$SSE_{(i)} = (SSE - e_i^2) - \frac{2h_{ii}e_i^2}{1 - h_{ii}} + \frac{h_{ii}e_i^2}{1 - h_{ii}} = SSE - \frac{e_i^2}{1 - h_{ii}}.$$

(c) $D_i = \frac{(\hat\beta - \hat\beta_{(i)})'A(\hat\beta - \hat\beta_{(i)})}{(p+1)s^2} = \frac{h_{ii}e_i^2}{(1-h_{ii})^2(p+1)s^2} = \frac{r_i^2}{p+1}\cdot\frac{h_{ii}}{1-h_{ii}}$.

Solution 18

(a) $y_i - \mathbf{x}_i'\hat\beta_{(i)} = e_i/(1 - h_{ii})$ is $N(0, \sigma^2/(1 - h_{ii}))$, because $1 + \mathbf{x}_i'(X_{(i)}'X_{(i)})^{-1}\mathbf{x}_i = 1/(1 - h_{ii})$. It is independent of $s_{(i)}^2$, which comes from the other observations, with $(n-p-2)s_{(i)}^2/\sigma^2 \sim \chi^2(n-p-2)$.

(b) Substitute $s^2_{(i)} = s^2(n-p-1-r_i^2)/(n-p-2)$ from Solution 17(b). Since $s^2_{(i)} \geq 0$, $r_i^2 \leq n-p-1$: the internally studentized residual is bounded, another sign that it is not t-distributed.

(c) The i-th leave-one-out prediction error is $y_i - \mathbf{x}_i'\hat{\beta}_{(i)} = e_i/(1-h_{ii})$ by Solution 17(a).

Solution 19

(a) $E\tilde{\beta}_1 = \frac{\sum(x_{i1}-\bar{x}_1)Ey_i}{S_{11}} = \beta_1 + \beta_2\frac{S_{12}}{S_{11}}$.

(b) $\operatorname{Var}\tilde{\beta}_1 = \sigma^2/S_{11}$, while $\operatorname{Var}\hat{\beta}_1 = \sigma^2/\{S_{11}(1-r_{12}^2)\}$. So dropping x_2 reduces the variance by the factor $1-r_{12}^2$.

(c) $\operatorname{MSE}(\tilde{\beta}_1) = \frac{\sigma^2}{S_{11}} + \beta_2^2\frac{S_{12}^2}{S_{11}^2}$ and $\operatorname{Var}\hat{\beta}_1 - \frac{\sigma^2}{S_{11}} = \frac{\sigma^2 r_{12}^2}{S_{11}(1-r_{12}^2)}$. Since $\operatorname{Var}\hat{\beta}_2 = \sigma^2/\{S_{22}(1-r_{12}^2)\}$ and $r_{12}^2 = S_{12}^2/(S_{11}S_{22})$, the inequality $\operatorname{MSE}(\tilde{\beta}_1) \leq \operatorname{Var}\hat{\beta}_1$ reduces to $\beta_2^2 \leq \operatorname{Var}\hat{\beta}_2$.

Solution 20

(a) Adding a regressor enlarges the column space, so the minimum SSE cannot increase.

(b) R_a^2 increases iff MSE decreases: $\frac{SSE_{k+1}}{n-k-2} < \frac{SSE_k}{n-k-1}$. With $F = \frac{SSE_k - SSE_{k+1}}{SSE_{k+1}/(n-k-2)}$ this is equivalent to $F > 1$.

(c) If the model is correct, $E(SSE_k) = (n-k-1)\sigma^2$, and with $\hat{\sigma}^2 \approx \sigma^2$, $E(C_k) \approx (n-k-1) + 2(k+1) - n = k+1$. For the full model $SSE_p/\hat{\sigma}^2 = n-p-1$ exactly, so $C_p = p+1$.

Solution 21

(a) $V = \operatorname{diag}(x_i)$, so GLS minimizes $\sum(y_i - \beta x_i)^2/x_i$: $\hat{\beta}_G = \sum y_i / \sum x_i$ (the ratio estimator), with $\operatorname{Var}\hat{\beta}_G = \sigma^2/\sum x_i$.

(b) $\operatorname{Var}\hat{\beta}_{OLS} = \sigma^2\sum x_i^3/(\sum x_i^2)^2$. By Cauchy–Schwarz, $(\sum x_i^2)^2 = (\sum x_i^{1/2}x_i^{3/2})^2 \leq \sum x_i \sum x_i^3$, so $\operatorname{Var}\hat{\beta}_{OLS} \geq \sigma^2/\sum x_i$.

(c) $\tilde{\sigma}^2 = \frac{1}{n-1}\sum(y_i - \hat{\beta}_G x_i)^2/x_i$.

Solution 22

(a) Use $(I+cJ)^{-1} = I - \frac{c}{1+nc}J$ (from $J^2 = nJ$) with $c = \rho/(1-\rho)$.

(b) $\mathbf{1} \in \operatorname{col}(X)$, and $V\mathbf{x} = (1-\rho)\mathbf{x} + \rho(\mathbf{1}'\mathbf{x})\mathbf{1} \in \operatorname{col}(X)$ for $\mathbf{x} \in \operatorname{col}(X)$. So $V^{-1}X = XB$ for some invertible B, and $(X'V^{-1}X)^{-1}X'V^{-1} = (B'X'X)^{-1}B'X' = (X'X)^{-1}X'$.

(c) $E\,\mathbf{y}'(I-H)\mathbf{y} = \sigma^2\operatorname{tr}\{(I-H)V\} = \sigma^2(1-\rho)(n-p-1)$, since $(I-H)J = 0$. So MSE estimates $(1-\rho)\sigma^2$, not σ^2, and the usual standard error of the intercept is wrong (the slopes' standard errors are correct when computed with $(1-\rho)\sigma^2$).

Solution 23

(a) $d_1 + d_2 + d_3 = \mathbf{1}$, the intercept column, so the columns are dependent and $X'X$ is singular.

(b) β_1 is the common slope; β_0 the intercept of the baseline level; β_2, β_3 the differences between levels 1, 2 and the baseline at any fixed x.

(c) $H_0 : \beta_2 = \beta_3 = 0$ by a partial F test with $(2, n-4)$ degrees of freedom.

Solution 24

(a) For $x \leq c$, $E(y) = \beta_0 + \beta_1 x$; for $x > c$, $(\beta_0 - \beta_2 c) + (\beta_1 + \beta_2)x$. Both equal $\beta_0 + \beta_1 c$ at c. The slopes are β_1 and $\beta_1 + \beta_2$.

(b) Add $\beta_3\mathbf{1}\{x > c\}$.

(c) t test of $H_0 : \beta_2 = 0$.

Solution 25

(a) The gradient of the penalized criterion is $-2X'(\mathbf{y} - X\beta) + 2k\beta$; it vanishes at $\hat{\beta}(k)$, and the

Hessian $2(X'X + kI)$ is positive definite. The augmented design gives normal equations $(X'X + kI)\beta = X'\mathbf{y}$.

(b) $E\hat{\beta}(k) = (X'X + kI)^{-1}X'X\beta$ and $\operatorname{Var}\hat{\beta}(k) = \sigma^2(X'X + kI)^{-1}X'X(X'X + kI)^{-1}$.

(c) In canonical coordinates, $g(k) = \sigma^2 \sum \frac{\lambda_j}{(\lambda_j+k)^2} + k^2 \sum \frac{\alpha_j^2}{(\lambda_j+k)^2}$ with $g(0) = E\|\hat{\beta} - \beta\|^2$ and $g'(0) = -2\sigma^2 \sum \lambda_j^{-2} < 0$.

(d) $g(k) = \frac{p\sigma^2 + k^2\|\beta\|^2}{(1+k)^2}$, and $g'(k) = 0$ gives $k = p\sigma^2/\|\beta\|^2$.

Solution 26

(a) By the partitioned-inverse formula, $c_{jj} = \{\mathbf{x}_j'(I - H_{-j})\mathbf{x}_j\}^{-1}$, the inverse of the residual sum of squares of x_j on the others, which is $1 - R_j^2$ because $\mathbf{x}_j'\mathbf{x}_j = 1$ for a standardized centred column.

(b) $\sum c_{jj} = \operatorname{tr}(X'X)^{-1} = \sum 1/\lambda_j$.

(c) VIF $= 1/(1 - 0.9025) = 10.26$, so standard errors (and interval widths) are $\sqrt{10.26} = 3.2$ times those for uncorrelated regressors.

Solution 27

(a) $E\tilde{\beta}_r = \sum_{k\le r} \mathbf{e}_k\mathbf{e}_k'\beta$, so the bias is $-\sum_{k>r} \mathbf{e}_k\mathbf{e}_k'\beta$; $\operatorname{Var}\tilde{\beta}_r = \sigma^2 \sum_{k\le r} l_k^{-1}\mathbf{e}_k\mathbf{e}_k'$.

(b) $E\|\tilde{\beta}_r - \beta\|^2 = \sigma^2 \sum_{k\le r} l_k^{-1} + \sum_{k>r} (\mathbf{e}_k'\beta)^2$, versus $\sigma^2 \sum_k l_k^{-1}$ for OLS; subtract.

(c) Canonical coefficient k is multiplied by 1 for $k \le r$ and 0 for $k > r$ in PCR, and by $l_k/(l_k + c)$ in ridge with constant c: PCR is hard thresholding, ridge smooth shrinkage.

Solution 28

(a) $l = \sum\{y_i\mathbf{x}_i'\beta - \log(1 + e^{\mathbf{x}_i'\beta})\}$, so the score is $X'(\mathbf{y} - \pi)$ and the Hessian is $-X'WX$ with $W = \operatorname{diag}\{\pi_i(1 - \pi_i)\}$, which is negative definite.

(b) $\beta^{new} = \beta + (X'WX)^{-1}X'(\mathbf{y} - \pi) = (X'WX)^{-1}X'W\mathbf{z}$ with $\mathbf{z} = X\beta + W^{-1}(\mathbf{y} - \pi)$: weighted least squares of $\mathbf{z}$ on X with weights W.

(c) e^{β_j} is the multiplicative change in the odds $P(y = 1)/P(y = 0)$ when x_j increases by one unit, the others fixed.

Solution 29

(a) The model is saturated, so the MLEs of $P(y = 1 \mid x)$ are the observed proportions n_{x1}/n_{x+}; then $e^{\hat{\beta}_1}$ is the ratio of the observed odds, $\frac{n_{11}/n_{10}}{n_{01}/n_{00}}$.

(b) Both odds ratios equal $\frac{p_{11}p_{00}}{p_{10}p_{01}}$ in terms of the cell probabilities. In a case–control study only $P(x \mid y)$ is estimable, yet it gives the same odds ratio, so the logistic slope is estimable.

(c) $\widehat{OR} = \frac{20\cdot 90}{80\cdot 10} = 2.25$, $\log\widehat{OR} = 0.811$, and se $= \sqrt{\frac{1}{20} + \frac{1}{80} + \frac{1}{10} + \frac{1}{90}} = 0.416$. The interval is $\exp(0.811 \pm 0.815) = (1.00, 5.08)$, borderline.

Solution 30

(a) Along $\beta = t(-2.5, 1)$ the linear predictors are $t(-1.5, -0.5, 0.5, 1.5)$, negative exactly for the zeros. Each term $y_i\eta_i - \log(1 + e^{\eta_i}) = -\log(1 + e^{-|\eta_i|})$ increases to 0 as $t \to \infty$.

(b) The supremum 0 is not attained by any finite β, since every term is < 0. So there is no MLE (complete separation).

(c) No: the least squares criterion is a quadratic with positive definite Hessian (full rank), so a minimizer always exists.

Solution 31

(a) $y_{jk} - \hat{y}_j = (y_{jk} - \bar{y}_j) + (\bar{y}_j - \hat{y}_j)$, and the cross term vanishes because $\sum_k (y_{jk} - \bar{y}_j) = 0$ within each group.

(b) $SS_{PE}/\sigma^2 \sim \chi^2(n-m)$ regardless of the mean function, and it depends only on within-group deviations, independent of the group means. If the line is correct, SS_{LOF} is the residual sum of squares of a weighted regression of the m group means, so $SS_{LOF}/\sigma^2 \sim \chi^2(m-2)$.

(c) Without replicates, $SS_{PE} = 0$ with 0 degrees of freedom: there is no model-free estimate of σ^2 to compare the lack of fit with.

Part II. Past exam problems (restated)

The regression problems of the SNU Statistics graduate entrance exams, October 2020 to April 2026, restated in our own words with the original mathematical content. (Quadratic-form problems appear in the mathematical statistics set.)

Past problem *[2020-10 #4]*

$\mathbf{y} \sim N(X\beta, V)$ with known nonsingular V. Let $\hat\beta = (X'V^{-1}X)^{-1}X'V^{-1}\mathbf{y}$, and let $\tilde\beta = A\mathbf{y}$ be any linear unbiased estimator. Show $\mathbf{a}'\{\operatorname{Var}\tilde\beta - \operatorname{Var}\hat\beta\}\mathbf{a} \geq 0$ for all $\mathbf{a}$.

Past problem *[2020-10 #5]*

$X = (X_1, X_2)$ of full rank ($n \times q$ and $n \times (p-q)$), H, H_1, H_2 the projections onto the column spaces of X, X_1, X_2, and $E(\mathbf{y} \mid X) = X_1\beta_1 + X_2\beta_2$.

(a) Express $\hat\beta_2$ using H_1, X_2, $\mathbf{y}$.

(b) Find $H\mathbf{y} - H_1\mathbf{y}$ when $X_1'X_2 = 0$.

Past problem *[2021-04 #1]*

Centred regression $y_i = \sum_j \beta_j x_{ij} + \epsilon_i$ with $\sum_i y_i = \sum_i x_{ij} = 0$, $n > p$, X of full rank.

(a) Write $SSR = \sum \hat y_i^2$ and $SST = \sum y_i^2$ in terms of X and $\mathbf{y}$.

(b) Prove $\max_{\beta \neq \mathbf{0}} \dfrac{\langle X\beta, \mathbf{y}\rangle^2}{\|X\beta\|^2\|\mathbf{y}\|^2} = \dfrac{SSR}{SST}$.

(c) Find the maximizer.

Past problem *[2021-04 #2]*

In the same centred model with uncorrelated errors of variance σ^2 but X possibly rank deficient, let $\hat\beta_{\text{ridge}}$ minimize $\|\mathbf{y} - X\beta\|^2 + k\|\beta\|^2$, $k > 0$.

(a) Find it.

(b) With λ_i the eigenvalues of $X'X$, prove $\sum_j \operatorname{Var}\hat\beta_{j,\text{ridge}} = \sigma^2 \sum_i \lambda_i/(\lambda_i + k)^2$.

Past problem *[2021-10 #1]*

$y_i = \mathbf{x}_i'\beta + e_i$, $i = 1, \ldots, n$, independent, $E(e_i \mid \mathbf{x}_i) = 0$, $\operatorname{Var}(e_i \mid \mathbf{x}_i) = \sigma^2$, and $\hat\beta$ is fitted on the first n_1 observations only.

(1) Express $E\frac{1}{n-n_1}\sum_{i>n_1}(y_i - \mathbf{x}_i'\hat\beta)^2$ in terms of σ^2 and the $\mathbf{x}_i$.

(2) Is the training error $\frac{1}{n_1}\sum_{i\leq n_1}(y_i - \mathbf{x}_i'\hat\beta)^2$ an unbiased estimator of it?

Past problem *[2021-10 #2]*

$E(\mathbf{y} \mid X) = X\beta$, $\operatorname{Var}(\mathbf{y} \mid X) = \sigma^2 I$, $X = UDV'$ (thin SVD). With $X^* = \binom{X}{\sqrt{\lambda} I_p}$ and $\mathbf{y}^* = \binom{\mathbf{y}}{\mathbf{0}_p}$, let $\hat\beta = (X^{*\prime}X^*)^{-1}X^{*\prime}\mathbf{y}^*$. Express $E(\hat\beta \mid X)$ and $\operatorname{Var}(\hat\beta \mid X)$ in terms of U, D, V, β, σ.

Past problem *[2022-04 #4]*

Define the singular value decomposition of an $n \times p$ matrix X ($n > p$) and the eigendecomposition of $X'X$, and relate them.

Past problem *[2022-04 #5]*

Describe how to diagnose high-leverage points, outliers and influential points.

Past problem *[2022-04 #6]*

Model $y_j = \mathbf{x}_{1j}'\beta_1 + \mathbf{x}_{2j}'\beta_2 + \epsilon_j$ ($\mathbf{x}_1$ includes the intercept). Regress y on $\mathbf{x}_1$ to get residuals e_j; regress each component of $\mathbf{x}_2$ on $\mathbf{x}_1$ to get residuals $\mathbf{x}_{2\cdot 1,j}$; regress e_j on $\mathbf{x}_{2\cdot 1,j}$ to get $\tilde\beta_2$.

(i) Prove that $\tilde{\beta}_2 = \hat{\beta}_2$ of the full model.

(ii) Interpret.

Past problem *[2023-04 #2]*

$\mathbf{y} = X\beta + \epsilon$, $E\epsilon = \mathbf{0}$, $\operatorname{Var}\epsilon = \sigma^2 V$ with V known positive definite, X of full rank $k+1$.

(1) Find the mean and covariance of $\tilde{\beta} = (X'X)^{-1}X'\mathbf{y}$.

(2) Show that the BLUE is $\hat{\beta} = (X'V^{-1}X)^{-1}X'V^{-1}\mathbf{y}$.

(3) Find $\operatorname{Var}\hat{\beta}$.

(4) Give an unbiased estimator of σ^2.

(5) How can V be estimated when unknown?

Past problem *[2023-10 #4]*

With $X = (X_k, X_r)$, let $\hat{\beta}_k$ be the part of the full least squares estimator and $\tilde{\beta}_k = (X_k'X_k)^{-1}X_k'\mathbf{y}$ the reduced-model estimator. Prove that $\operatorname{Var}(\hat{\beta}_k) - \operatorname{Var}(\tilde{\beta}_k)$ is nonnegative definite.

Past problem *[2023-10 #5]*

$y_i = \theta_i + \epsilon_i$, $i = 1, \dots, 4$, $\epsilon_i \overset{\text{iid}}{\sim} N(0, \sigma^2)$, with the constraint $\theta_1 + \theta_2 + \theta_3 + \theta_4 = 0$. Show that the F statistic for $H_0 : \theta_1 = \theta_3$ is $F_0 = \dfrac{2(y_1 - y_3)^2}{(y_1 + y_2 + y_3 + y_4)^2}$.

Past problem *[2024-05 #5]*

Two-way ANOVA without replication, $y_{ij} = \mu + \alpha_i + \beta_j + \epsilon_{ij}$, $i \le a$, $j \le b$, $\sum \alpha_i = \sum \beta_j = 0$, $\epsilon_{ij} \overset{\text{iid}}{\sim} N(0, \sigma^2)$.

(1) Prove that $Q_1 = \sum_{i,j}(\bar{y}_{i\cdot} - \bar{y}_{\cdot\cdot})^2$ and $Q_2 = \sum_{i,j}(y_{ij} - \bar{y}_{i\cdot} - \bar{y}_{\cdot j} + \bar{y}_{\cdot\cdot})^2$ are independent.

(2) Find an unbiased estimator of σ^2 and a $100(1-\alpha)\%$ confidence interval for $\mu + \alpha_1 + \beta_1$.

Past problem *[2024-10 #4]*

$(\mathbf{x}_i, y_i) \in \mathbb{R}^p \times \mathbb{R}$, $\sum \mathbf{x}_i \mathbf{x}_i'$ positive definite, $\hat{\beta}$ the least squares estimate.

(a) If $y_i = x_{ij}$ for all i and a fixed j, find $\hat{\beta}$.

(b) Assuming $\sum y_i^2 > 0$, prove $\max_\beta \dfrac{(\sum y_i \mathbf{x}_i'\beta)^2}{\sum y_i^2 \sum (\mathbf{x}_i'\beta)^2} = \dfrac{\sum \hat{y}_i^2}{\sum y_i^2}$, and evaluate it in case (a).

Past problem *[2024-10 #5]*

For convex $L : \mathbb{R} \to \mathbb{R}$, $\partial L(\theta) = \{z : L(\phi) \ge L(\theta) + z(\phi - \theta)\ \forall \phi\}$.

(a) Prove θ minimizes L iff $0 \in \partial L(\theta)$.

(b) Show $\partial|\theta| = \{\operatorname{sgn}\theta\}$ for $\theta \neq 0$ and $[-1, 1]$ at 0. (c), (d) For centred data with $\sum x_i^2 > 0$, let $L_\lambda(\theta) = \frac{1}{2}\sum (y_i - x_i\theta)^2 + \lambda|\theta|$ and θ_0 the least squares slope. Prove the minimizer is $\hat{\theta}_\lambda = \operatorname{sgn}(\theta_0)\big(|\theta_0| - \lambda / \sum x_i^2\big)_+$. Is it unique?

Past problem *[2025-04 #4]*

For $\mathbf{y} = X\beta + \epsilon$ with k regressors and intercept:

(1) find $E(\mathbf{e}'\mathbf{e})$ and $\operatorname{Cov}(\hat{\beta}, \mathbf{e})$;

(2) show $\sum_i \operatorname{Var}(\hat{y}_i) = c\sigma^2$ and find c;

(3) show $H - \mathbf{1}(\mathbf{1}'\mathbf{1})^{-1}\mathbf{1}'$ is idempotent.

Past problem *[2025-04 #5]*

$\mathbf{y} = X_1\beta_1 + X_2\beta_2 + \epsilon$. Show that $R^2(F) \ge R^2(R)$, where R is the reduced model under $H_0 : \beta_2 = \mathbf{0}$.

Past problem *[2025-04 #6]*

For a submodel with p parameters, $C_p = SSE(p)/\hat{\sigma}^2 - n + 2p$, $SSE(p) = \mathbf{y}'(I - H_p)\mathbf{y}$. Show $E(C_p) \approx p$ if $\mu = E\mathbf{y} = X_p\beta_p$.

Past problem *[2025-10 #4]*

$y_i = \beta_0 + \beta_1 x_i + \epsilon_i$ with independent $\epsilon_i \sim N(0, \sigma_i^2)$.

(a) If $\sigma_i^2 = \sigma^2$, find the MLEs of β_0, β_1.

(b) If $\sigma_i^2 = \sigma_1^2$ for $i \le n_1$ and σ_2^2 for $i > n_1$, both unknown, find the MLEs of $\beta_0, \beta_1, \sigma_1^2, \sigma_2^2$.

(c) Find the LSE and the WLSE.

(d) Are the MLE and WLSE identical?

Past problem *[2025-10 #5]*

Ridge $\hat{\beta}_R = (X'X + \lambda I)^{-1}X'\mathbf{y}$.

(a) Find its mean and covariance.

(b) Show that for some λ its variance is smaller than that of the LSE.

(c) Find and interpret the limits as $\lambda \to 0$ and $\lambda \to \infty$.

Past problem *[2026-04 #4]*

For $\mathbf{y} = X\beta + \epsilon$ and full-rank C ($(p+1) \times q$):

(a) sketch why the minimizer of $\|\mathbf{y} - X\beta\|^2$ subject to $C'\beta = \mathbf{0}$ is $\hat{\beta}_r = \hat{\beta} - (X'X)^{-1}C\{C'(X'X)^{-1}C\}^{-1}C'\hat{\beta}$;

(b) describe the partial F test of $C'\beta = \mathbf{0}$ and its null distribution;

(c) derive the distribution of $C'\hat{\beta}$ and the corresponding F test;

(d) are the tests in (b) and (c) equivalent?

Past problem *[2026-04 #5]*

(a) Show $e_i/(\sigma\sqrt{1 - h_{ii}}) \sim N(0, 1)$.

(b) Why is $r_i = e_i/(s\sqrt{1 - h_{ii}})$ not t-distributed? (c) With $s_{(i)}$ computed without observation i and $r_i^* = e_i/(s_{(i)}\sqrt{1 - h_{ii}})$, show (i) $s_{(i)}^2 = \{(n - p - 1)s^2 - e_i^2/(1 - h_{ii})\}/(n - p - 2)$ and (ii) $\hat{\beta}_{(i)} = \hat{\beta} - (X'X)^{-1}\mathbf{x}_i e_i/(1 - h_{ii})$.

Part II. Solutions

Solution 2020-10 #4

Unbiasedness gives $AX = I$. Write $A = (X'V^{-1}X)^{-1}X'V^{-1} + D$, so $DX = 0$. Then

$$\operatorname{Var}\tilde{\beta} = AVA' = (X'V^{-1}X)^{-1} + DVD' + (X'V^{-1}X)^{-1}X'D' + DX(X'V^{-1}X)^{-1},$$

and the cross terms vanish because $DX = 0$. Hence $\mathbf{a}'(\operatorname{Var}\tilde{\beta} - \operatorname{Var}\hat{\beta})\mathbf{a} = \mathbf{a}'DVD'\mathbf{a} = \|V^{1/2}D'\mathbf{a}\|^2 \geq 0$.

Solution 2020-10 #5

(a) $\hat{\beta}_2 = \{X_2'(I - H_1)X_2\}^{-1}X_2'(I - H_1)\mathbf{y}$ (Part I, Problem 12).

(b) If $X_1'X_2 = 0$, the column spaces are orthogonal and $H = H_1 + H_2$, so $H\mathbf{y} - H_1\mathbf{y} = H_2\mathbf{y} = X_2(X_2'X_2)^{-1}X_2'\mathbf{y}$.

Solution 2021-04 #1

(a) $SSR = \mathbf{y}'X(X'X)^{-1}X'\mathbf{y} = \mathbf{y}'H\mathbf{y}$ and $SST = \mathbf{y}'\mathbf{y}$ (centred data).

(b) $\langle X\beta, \mathbf{y}\rangle = \langle X\beta, H\mathbf{y}\rangle$ because $X\beta$ lies in the column space. By Cauchy–Schwarz, $\langle X\beta, H\mathbf{y}\rangle^2 \leq \|X\beta\|^2\|H\mathbf{y}\|^2$, so the ratio is at most $\|H\mathbf{y}\|^2/\|\mathbf{y}\|^2 = SSR/SST$, with equality when $X\beta \propto H\mathbf{y}$.

(c) $\beta = c\hat{\beta}$ for any $c \neq 0$: the maximum squared correlation between $\mathbf{y}$ and a linear combination of the regressors is attained by the fitted values (the multiple correlation).

Solution 2021-04 #2

(a) $\hat{\beta}_{\text{ridge}} = (X'X + kI)^{-1}X'\mathbf{y}$; $X'X + kI$ is positive definite even if X is rank deficient.

(b) $\operatorname{Var} = \sigma^2(X'X + kI)^{-1}X'X(X'X + kI)^{-1} = \sigma^2 P\,\operatorname{diag}\{\lambda_i/(\lambda_i + k)^2\}P'$ with $X'X = P\Lambda P'$. Take the trace.

Solution 2021-10 #1

(1) Let X_1 be the training design. For $i > n_1$, y_i is independent of $\hat{\beta}$, and $y_i - \mathbf{x}_i'\hat{\beta}$ has mean 0 and variance $\sigma^2\{1 + \mathbf{x}_i'(X_1'X_1)^{-1}\mathbf{x}_i\}$. So the expectation is

$$\sigma^2 + \frac{\sigma^2}{n - n_1}\sum_{i>n_1}\mathbf{x}_i'(X_1'X_1)^{-1}\mathbf{x}_i.$$

(2) $E\frac{1}{n_1}\sum_{i\leq n_1}(y_i - \mathbf{x}_i'\hat{\beta})^2 = \frac{n_1 - p}{n_1}\sigma^2 < \sigma^2$, which is below (1). So the training error is biased downward and is not unbiased; it underestimates the prediction error.

Solution 2021-10 #2

$X^{*\prime}X^* = X'X + \lambda I$ and $X^{*\prime}\mathbf{y}^* = X'\mathbf{y}$, so $\hat{\beta}$ is the ridge estimator. With $X = UDV'$,

$$\hat{\beta} = V(D^2 + \lambda I)^{-1}DU'\mathbf{y},$$

$$E(\hat{\beta} \mid X) = V(D^2 + \lambda I)^{-1}D^2V'\beta, \qquad \operatorname{Var}(\hat{\beta} \mid X) = \sigma^2 V(D^2 + \lambda I)^{-2}D^2V'.$$

Solution 2022-04 #4

$X = UDV'$ with U ($n \times p$, orthonormal columns), $D = \operatorname{diag}(d_1 \geq \cdots \geq d_p \geq 0)$ the singular values, and V ($p \times p$ orthogonal). Then $X'X = VD^2V'$: the eigenvectors of $X'X$ are the right singular vectors and the eigenvalues are d_j^2. Also $XV = UD$: the principal components $X\mathbf{v}_j = d_j\mathbf{u}_j$.

Solution 2022-04 #5

Leverage: $h_{ii} = \mathbf{x}_i'(X'X)^{-1}\mathbf{x}_i$, equivalently the Mahalanobis distance of $\mathbf{x}_i$; flag $h_{ii} > 2(p + 1)/n$. *Outliers*: studentized residuals $r_i^* = e_i/(s_{(i)}\sqrt{1 - h_{ii}}) \sim t(n - p - 2)$; flag large $|r_i^*|$, with a

Bonferroni correction. *Influence*: Cook's distance $D_i = \frac{r_i^2}{p+1}\frac{h_{ii}}{1-h_{ii}}$ (compare with $F_{0.5}(p+1, n-p-1)$), DFFITS$_i$ (flag $> 2\sqrt{(p+1)/n}$), DFBETAS and COVRATIO. Then inspect flagged cases for errors, and refit with and without them.

Solution 2022-04 #6

(i) Part I, Problem 12(a)–(b) (Frisch–Waugh–Lovell).

(ii) A multiple regression coefficient is the effect of x_2 on y after both have been purged of their linear association with x_1: the partial effect holding x_1 fixed.

Solution 2023-04 #2

(1) $E\tilde{\beta} = \beta$ and $\operatorname{Var}\tilde{\beta} = \sigma^2(X'X)^{-1}X'VX(X'X)^{-1}$.

(2) Transform by $V^{-1/2}$ to obtain Var $= \sigma^2 I$, then apply Gauss–Markov; or see Past problem 2020-10 #4.

(3) $\sigma^2(X'V^{-1}X)^{-1}$.

(4) $\tilde{\sigma}^2 = (\mathbf{y} - X\hat{\beta})'V^{-1}(\mathbf{y} - X\hat{\beta})/(n-k-1)$.

(5) Parametrize $V = V(\theta)$ (AR(1), equicorrelation, variance functions), estimate θ from OLS residuals, and plug in (feasible GLS), possibly iterating.

Solution 2023-10 #4

With $A = (X_k'X_k)^{-1}X_k'X_r$ and $B = X_r'(I - H_k)X_r$, the partitioned inverse gives $\operatorname{Var}\hat{\beta}_k = \sigma^2\{(X_k'X_k)^{-1} + AB^{-1}A'\}$, while $\operatorname{Var}\tilde{\beta}_k = \sigma^2(X_k'X_k)^{-1}$. The difference $\sigma^2 AB^{-1}A'$ is nonnegative definite.

Solution 2023-10 #5

The full parameter space $\{\theta : \mathbf{1}'\theta = 0\}$ has dimension 3, so $SSE(F) = \|\mathbf{y} - \hat{\theta}\|^2$ is the squared projection of $\mathbf{y}$ on $\mathbf{1}$: $(\sum y_i)^2/4$, with 1 degree of freedom. Under H_0 the space is $\{\mathbf{1}'\theta = 0,\ \theta_1 = \theta_3\}$, whose orthogonal complement is spanned by the orthogonal vectors $\mathbf{1}$ and $(1, 0, -1, 0)'$. So $SSE(R) = (\sum y_i)^2/4 + (y_1 - y_3)^2/2$. Hence

$$F_0 = \frac{\{SSE(R) - SSE(F)\}/1}{SSE(F)/1} = \frac{2(y_1 - y_3)^2}{(y_1 + y_2 + y_3 + y_4)^2} \sim F(1,1) \text{ under } H_0.$$

Solution 2024-05 #5

(1) Q_1 is a function of the row means $\bar{y}_{i\cdot}$, and Q_2 is a function of the interaction residuals $r_{ij} = y_{ij} - \bar{y}_{i\cdot} - \bar{y}_{\cdot j} + \bar{y}_{\cdot\cdot}$. These are jointly normal and uncorrelated: $\operatorname{Cov}(\bar{y}_{i\cdot}, r_{kl}) = \frac{\sigma^2}{b}\{\delta_{ik} - \frac{1}{a} - \delta_{ik} + \frac{1}{a}\} = 0$ (row averaging kills the column-centred residual). Hence they are independent. Equivalently, the projection matrices P_A and P_{AB} satisfy $P_A P_{AB} = 0$.

(2) $Q_2/\sigma^2 \sim \chi^2((a-1)(b-1))$, so $\hat{\sigma}^2 = Q_2/\{(a-1)(b-1)\}$. The estimator $\bar{y}_{1\cdot} + \bar{y}_{\cdot 1} - \bar{y}_{\cdot\cdot}$ of $\mu + \alpha_1 + \beta_1$ has variance

$$\sigma^2\Big(\frac{1}{b} + \frac{1}{a} - \frac{1}{ab}\Big) = \sigma^2\frac{a+b-1}{ab},$$

so the interval is $\bar{y}_{1\cdot} + \bar{y}_{\cdot 1} - \bar{y}_{\cdot\cdot} \pm t_{\alpha/2}((a-1)(b-1))\sqrt{\hat{\sigma}^2(a+b-1)/(ab)}$.

Solution 2024-10 #4

(a) $\mathbf{y} = X\mathbf{e}_j$ lies in the column space, so $\hat{\beta} = \mathbf{e}_j$.

(b) As in Past problem 2021-04 #1 (without centring): the maximum is $\|H\mathbf{y}\|^2/\|\mathbf{y}\|^2 = \sum \hat{y}_i^2 / \sum y_i^2$. In case (a) it equals 1.

Solution 2024-10 #5

(a) $0 \in \partial L(\theta)$ iff $L(\phi) \geq L(\theta)$ for all ϕ.

(b) For $\theta > 0$, the inequality $|\phi| \geq \theta + z(\phi - \theta)$ for ϕ near θ forces $z = 1$; similarly for $\theta < 0$. At 0, $|\phi| \geq z\phi$ for all ϕ iff $|z| \leq 1$.

(c), (d) With $S = \sum x_i^2$ and centred data, $L_\lambda(\theta) = \frac{S}{2}(\theta - \theta_0)^2 + \lambda|\theta| + \text{const}$. So $\partial L_\lambda(\theta) = S(\theta - \theta_0) + \lambda\partial|\theta|$. For $\theta > 0$ we need $\theta = \theta_0 - \lambda/S > 0$; for $\theta < 0$, $\theta = \theta_0 + \lambda/S < 0$; and $\theta = 0$ works iff $|S\theta_0| \leq \lambda$. Together this is $\hat{\theta}_\lambda = \operatorname{sgn}(\theta_0)(|\theta_0| - \lambda/S)_+$ (soft thresholding). It is unique because L_λ is strictly convex ($S > 0$).

Solution 2025-04 #4

(1) $E\mathbf{e}'\mathbf{e} = \sigma^2 \operatorname{tr}(I - H) = (n - k - 1)\sigma^2$, and $\operatorname{Cov}(\hat{\beta}, \mathbf{e}) = (X'X)^{-1}X'\sigma^2(I - H) = 0$.

(2) $\operatorname{Var}\hat{\mathbf{y}} = \sigma^2 H$, so $\sum \operatorname{Var}\hat{y}_i = \sigma^2 \operatorname{tr} H$ and $c = k + 1$.

(3) $HJ = JH = J$ (since $H\mathbf{1} = \mathbf{1}$), so $(H - J/n)^2 = H - 2J/n + J/n = H - J/n$.

Solution 2025-04 #5

SST is the same for both models, and $SSE(F) \leq SSE(R)$ because the reduced model minimizes over a subspace. Hence $R^2(F) = 1 - SSE(F)/SST \geq 1 - SSE(R)/SST = R^2(R)$.

Solution 2025-04 #6

If $\mu = X_p\beta_p$, then $(I - H_p)\mu = \mathbf{0}$ and $E\,SSE(p) = \sigma^2 \operatorname{tr}(I - H_p) = (n - p)\sigma^2$. With $\hat{\sigma}^2$ (from the full model) approximately σ^2, $E(C_p) \approx (n - p) - n + 2p = p$.

Solution 2025-10 #4

(a) OLS: $\hat{\beta}_1 = S_{xy}/S_{xx}$, $\hat{\beta}_0 = \bar{y} - \hat{\beta}_1\bar{x}$.

(b) The log-likelihood is $-\frac{n_1}{2}\log\sigma_1^2 - \frac{n_2}{2}\log\sigma_2^2 - \frac{SS_1(\beta)}{2\sigma_1^2} - \frac{SS_2(\beta)}{2\sigma_2^2}$, with SS_g the group residual sums of squares. The MLE solves jointly $\hat{\sigma}_g^2 = SS_g(\hat{\beta})/n_g$ and $\hat{\beta}$ = the WLS estimator with weights $1/\hat{\sigma}_g^2$, computed iteratively; there is no closed form in general.

(c) The LSE is OLS as in (a). The WLSE is $(X'WX)^{-1}X'W\mathbf{y}$ with $W = \operatorname{diag}(1/\sigma_g^2)$, in practice with σ_g^2 estimated by group residual mean squares from an OLS fit.

(d) The MLE is a fixed point of iterated WLS with the variances estimated by SS_g/n_g. A one-step WLSE using OLS-based variance estimates (or divisors $n_g - 2$) differs from the MLE in general; they coincide when the WLS is iterated to convergence with the ML variance estimates.

Solution 2025-10 #5

(a) $E\hat{\beta}_R = (X'X + \lambda I)^{-1}X'X\beta$ and $\operatorname{Var}\hat{\beta}_R = \sigma^2(X'X + \lambda I)^{-1}X'X(X'X + \lambda I)^{-1}$.

(b) With $X'X = P\Lambda P'$, $\operatorname{Var}\hat{\beta} - \operatorname{Var}\hat{\beta}_R = \sigma^2 P \operatorname{diag}\left\{\frac{1}{\lambda_j} - \frac{\lambda_j}{(\lambda_j+\lambda)^2}\right\}P'$, and each entry is positive for every $\lambda > 0$: the variance is always smaller (and the mean squared error is smaller for small λ).

(c) $\lambda \to 0$: $\hat{\beta}_R \to \hat{\beta}$ (OLS). $\lambda \to \infty$: $\hat{\beta}_R \approx X'\mathbf{y}/\lambda \to \mathbf{0}$ (all coefficients shrunk to zero; zero variance, maximal bias).

Solution 2026-04 #4

(a) Lagrange: $X'X\beta + C\mu = X'\mathbf{y}$ and $C'\beta = \mathbf{0}$ give $\mu = \{C'(X'X)^{-1}C\}^{-1}C'\hat{\beta}$ and the stated $\hat{\beta}_r$.

(b) Fit the full and the restricted model. Then $F = \frac{\{SSE(R) - SSE(F)\}/q}{SSE(F)/(n-p-1)} \sim F(q, n-p-1)$ under H_0, because $SSE(R) - SSE(F) = \mathbf{y}'(H - H_R)\mathbf{y}$ with $H - H_R$ idempotent of rank q and orthogonal to $I - H$.

(c) $C'\hat{\beta} \sim N(C'\beta, \sigma^2 C'(X'X)^{-1}C)$, so $F = \frac{(C'\hat{\beta})'\{C'(X'X)^{-1}C\}^{-1}C'\hat{\beta}/q}{MSE} \sim F(q, n-p-1)$ under H_0.

(d) Yes: $SSE(R) - SSE(F) = (\hat{\beta} - \hat{\beta}_r)'X'X(\hat{\beta} - \hat{\beta}_r) = (C'\hat{\beta})'\{C'(X'X)^{-1}C\}^{-1}C'\hat{\beta}$, so the statistics are identical.

Solution 2026-04 #5

(a) $\mathbf{e} = (I - H)\mathbf{y} \sim N(\mathbf{0}, \sigma^2(I - H))$, so $e_i \sim N(0, \sigma^2(1 - h_{ii}))$.

(b) e_i and s are dependent (e_i^2 is part of SSE), so the ratio is not a normal over an independent chi; indeed $r_i^2 \le n - p - 1$ is bounded.

(c) Part I, Problem 17(a)–(b).